National Institute of Technology Goa

Electrical and Electronics Department

Analog and Digital Circuits Lab

Mini project

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DIGITAL IC TESTER FOR LOGIC GATES

1. Aim

To design and implement a digital IC tester capable of verifying the functionality of basic logic gate ICs such as AND, OR, NAND, XOR, and XNOR gates.

2. Apparatus / Components Required

Sl. No	Component	Specification	Quantity
1	ICs under test	AND (7408), OR (7432), NAND (7400), XOR (7486), XNOR (74266/74286)	1 each
2	Timer IC	555 Timer	1
3	Flip-Flop IC	JK Flip-Flop (7476)	1
4	Multiplexer	4:1 MUX (74153)	1
5	XOR Gate IC	7486	1
6	LEDs	Red/Green indicators	4–6
7	Resistors	220Ω, 1kΩ, 12kΩ	As required
8	Capacitors	0.01µF, 10µF	As required
9	Power Supply	+5V DC regulated	1
10	Breadboard / PCB	Standard	1
11	Connecting wires	—	As required

3)Theory

3.1 555 Timer as Astable Multivibrator

The 555 timer IC is configured in astable mode to generate continuous square wave pulses without any external triggering.

- It produces a clock signal (0 and 1 pulses).
- These pulses are used to drive the JK flip-flop counter.
- Frequency depends on resistor and capacitor values:

$$f = \frac{1.44}{(R_1 + 2R_2)C}$$

3.2 JK Flip-Flop as Counter

The JK flip-flop is used in toggle mode (J=K=1) to create a binary counter.

- Each clock pulse toggles the output.
- A 2-bit counter generates input combinations:
  - 00, 01, 10, 11
- These combinations are used as test inputs for logic gates.

3.3 Multiplexer (MUX) for Input Selection

A 4:1 multiplexer selects the required input combination for testing.

- Select lines are controlled by the counter outputs.
- It routes specific inputs to the IC under test.
- Helps in automating testing without manual switchin

3.4 XOR Gate for Output Verification

The XOR gate is used as a comparator:

- One input: Output of the IC under test
- Second input: Expected output (reference logic)
- If both outputs are same → XOR output = 0 → IC is working correctly
- If outputs differ → XOR output = 1 → IC is faulty

LED indication:

- OFF → Correct
- ON → Fault detected

Note

For testing NOT Gate and NOR Gate we need small modification in the circuit

NOT gate uses only one input, so one input line is ignored and the output is taken as the complement of the input.

NOR gate requires wiring modification because its input and output pins are arranged differently from other logic gate ICs.

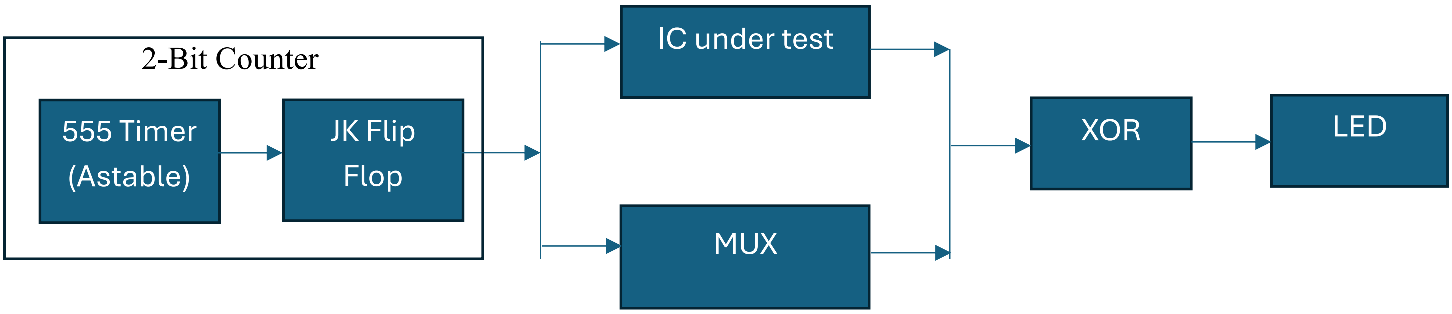

Diagram 1:Block diagram of working

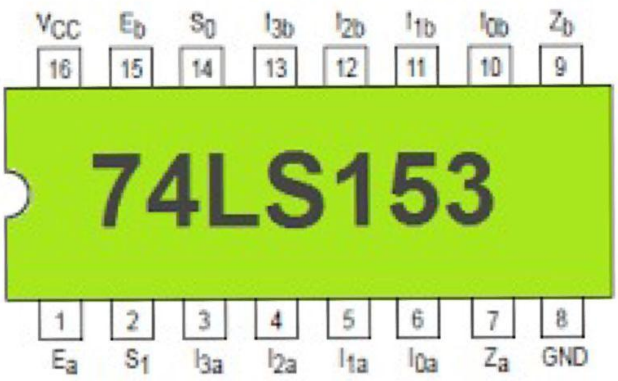

Fig 1:pin diagram of 4:1 MUX

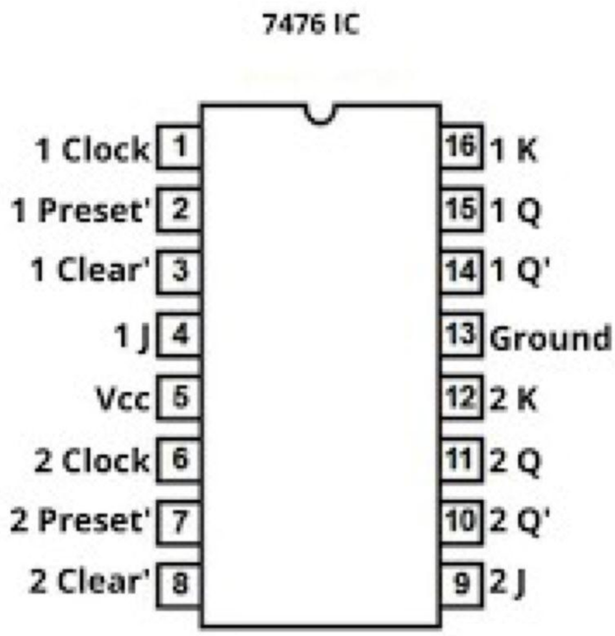

Fig 2:pin diagram of JK Flipflop

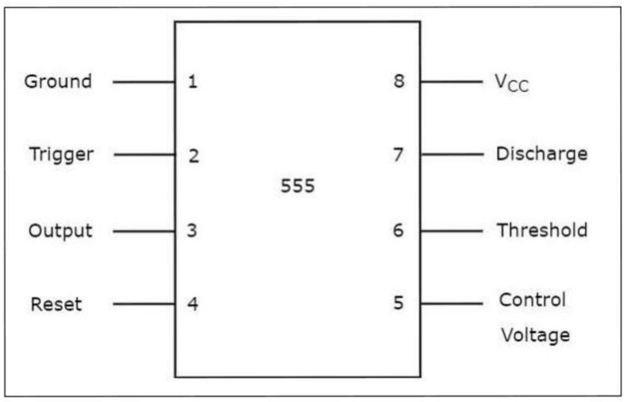

Fig 3:pin diagram of 555 Timer

Circuit diagram:

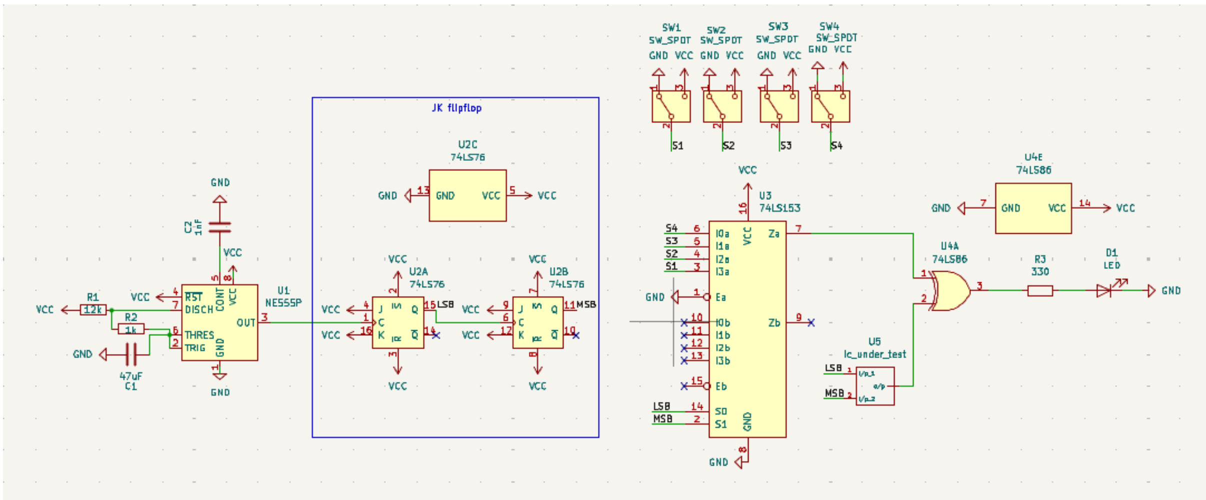

Fig 3:circuit diagram